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Hiring Team
Sanofi Manufacturing & Supply
Lyon, France

Subject: Application for the Data Scientist position in Lyon

Dear Hiring Team,

I am applying for the Data Scientist position within Sanofi Manufacturing & Supply in Lyon. I hold a PhD in experimental physics and have several years of postdoctoral experience developing Python/C++ workflows for statistical modelling, machine-learning validation, calibration, monitoring, and inference on complex measurement data. The opportunity to apply these methods to real manufacturing and laboratory data, and to build robust digital products with scientists, engineers, and MLOps colleagues, is closely aligned with the kind of problem solving I want to pursue in industry.

My work has repeatedly involved situations where observations were noisy, direct ground truth was incomplete, and model outputs could not be trusted without careful validation. I implemented likelihood-based and time-dependent models, developed CatBoost pipelines with train/test diagnostics, aligned simulated and observed datasets through reweighting and control-sample calibration, and built monitoring workflows that traced discrepancies to physical-system or software causes. These analyses produced concrete outcomes, including calibration updates, reconstruction fixes, an approximately twofold improvement in reconstructed-energy resolution in a representative study, and a quantified response non-uniformity contribution below 1% in a measurement-to-model validation study.

I also bring experience working in Linux-based shared codebases and large international collaborations, where reproducibility, technical review, and clear communication were essential. While my background is in scientific research rather than pharmaceutical manufacturing, the transferable core is robust: organising complex data, testing model assumptions, quantifying uncertainty, diagnosing failures, and turning analysis into defensible decisions. I would be transparent that I have not yet owned cloud deployment or a production MLOps lifecycle; I am actively prepared to develop those capabilities and would contribute immediately on modelling, validation, monitoring, and scientific problem framing.

My decision to move from academia to industry is deliberate. I have become clear that what motivates me most is solving difficult, real-world problems rather than particle physics itself. Sanofi's combination of advanced manufacturing, real-time equipment data, cross-functional product development, and direct patient impact offers a particularly meaningful setting for that transition. I work professionally in English and am continuing to strengthen my French, currently at B1-B2 level.

I would welcome the opportunity to discuss how my experience in statistical modelling, model credibility, and complex measurement data could support Sanofi's Digital M&S Accelerator.

Yours sincerely,

Yingrui Hou